

The enteric nervous system (ENS), which consists of two subdivisions known as the submucosal nerve plexus and the myenteric nerve plexus, innervates and regulates functions of the gastrointestinal (GI) tract. Recent studies have focused on the involvement of the ENS in Parkinson's disease (PD), a disorder characterized by motor dysfunction.

A form of PD associated with late-onset symptoms is caused by a mutation in a gene encoding the protein α -synuclein that leads to the misfolding of α -synuclein proteins. These mutant proteins generally aggregate into clusters referred to as Lewy bodies, which accumulate in the central nervous system (CNS) over time, impairing motor neuron function (eg, reduced secretion of certain proteins and neurotransmitters) and eventually causing neuron death.

Scientists hypothesize that Lewy bodies in individuals with this mutation may originate in the ENS prior to the onset of PD symptoms. This hypothesis was tested in experiments using 2-month-old wild-type mice (WT mice) and 2-month-old mice with this α -synuclein mutation (MT mice) exhibiting no PD motor symptoms.

Experiment 1

For both WT and MT mice, muscle contractions and the speed at which food moved along the tract were measured in the small intestine (Figure 1).

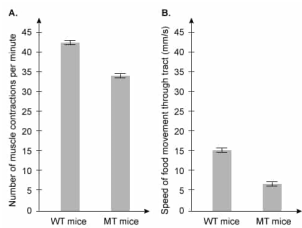


Figure 1 Number of muscle contractions (A) and speed of material movement through the tract (B) in small intestines of WT and MT mice

Experiment 2

Equal numbers of cells (approximately 400) from the myenteric nerve plexus were isolated from WT mice and exposed to either a control solution lacking α -synuclein, 0.5 μ M monomeric (nonaggregated) mutant α -synuclein proteins, or 0.5 μ M aggregated mutant α -synuclein proteins. After 5 days, the total numbers of living cells were determined along with the numbers of these living cells expressing a protein called PGP9.5 (Figure 2), which is known to be expressed in healthy neurons.

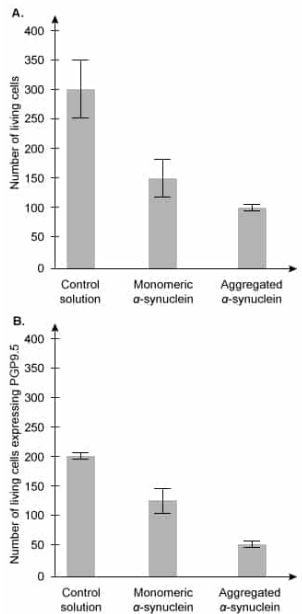


Figure 2 Number of living cells (A) and number of living cells expressing PGP9.5 (B) 5 days after exposure to either a control solution lacking α -synuclein, 0.5 μ M monomeric (nonaggregated) mutant α -synuclein proteins, or 0.5 μ M aggregated mutant α -synuclein proteins

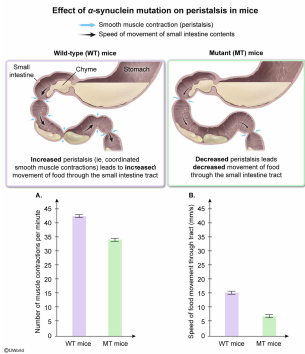
Which of the following conclusions regarding small intestine function in MT mice compared to WT mice is best supported by the information shown in Figure 1?

- ☐ A. MT mice increased their rate of muscle contraction to compensate for reduced movement of food through the small intestine tract. [2%]
- ☐ B. MT mice exhibited decreased rates of sarcomere contractions in muscle surrounding the small intestine. [39%]
- ☒ C. MT mice exhibited reduced coordination of nonstriated muscle contractions in the small intestine tract lining. [56%]
- ☐ D. MT mice more rapidly accumulated waste from the small intestine in the colon. [1%]

Correct

56% answered correctly

Explanation:



The components of the **gastrointestinal (GI) tract** include the mouth, esophagus, **stomach**, **small intestine**, and **large intestine**. Masticated food from the mouth travels down the esophagus, entering the stomach where it is converted into chyme (ie, a semifluid partially digested food mixture). Next, the chyme enters the small intestine, where further digestion of **macromolecules** occurs along with absorption of water and nutrients. Next, absorption of remaining water and electrolyte molecules (eg, sodium, chloride) occurs in the large intestine. Any remaining waste material in the large intestine is eliminated as feces through the rectum.

The movement of food along the GI tract occurs via **peristalsis**, the coordinated contraction of **smooth (nonstriated) muscle** tissue lining the intestine walls.

In this passage, small intestine function was analyzed in 2-month-old wild-type (WT) mice and 2-month-old mice with a **mutation** (ie, MT mice) in the gene encoding the α -synuclein protein. This mutation is known to lead to the development of Parkinson's disease (PD). Figure 1 shows that, compared to WT mice, MT mice exhibited **fewer** muscle contractions per minute in the small intestine as well as **decreased** speed of food movement through the small intestine tract.

Because **smooth (nonstriated) muscle** contraction (ie, peristalsis) controls the movement of food through the small intestine portion of the GI tract, these data indicate that, compared to WT mice, MT mice likely exhibited **reduced coordination of nonstriated muscle contractions** in the small intestine tract lining.

(Choice A) Although Figure 1 shows that MT mice exhibited reduced speed of food movement through the small intestine tract, the rate of muscle contraction (ie, number of contractions per minute) in MT mice decreased (*not* increased) compared to WT mice.

(Choice B) Skeletal muscle and cardiac muscle are striated and contain **sarcomeres**, the functional units crucial to muscle **contraction**. However, smooth muscle (ie, intestinal muscle) *lacks* sarcomeres, so MT mice would *not* have exhibited decreased rates of sarcomere contraction in muscle surrounding the small intestine.

(Choice D) Because Figure 1 shows that food moved more *slowly* through the small intestines of the MT mice, materials were also likely to move more slowly from the small intestines into the large intestines. This likely led to less (*not* more) rapid waste accumulation in the colons (ie, portion of large intestine) of MT mice.

Educational objective:

Peristalsis is the coordinated smooth (nonstriated) muscle contraction that propels food along the gastrointestinal tract, which comprises the mouth, esophagus, stomach, small intestine, and large intestine.

Time spent:0 sec
Subject:
Biology

QID:403225
System:
Digestion and Excretion

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Experiment 1

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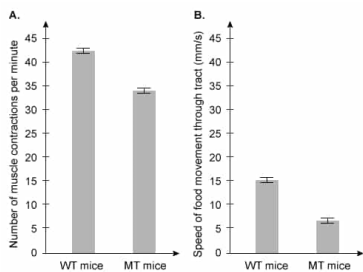


Figure 1 Number of muscle contractions (A) and speed of material movement through the tract (B) in small intestines of WT and MT mice

Experiment 2

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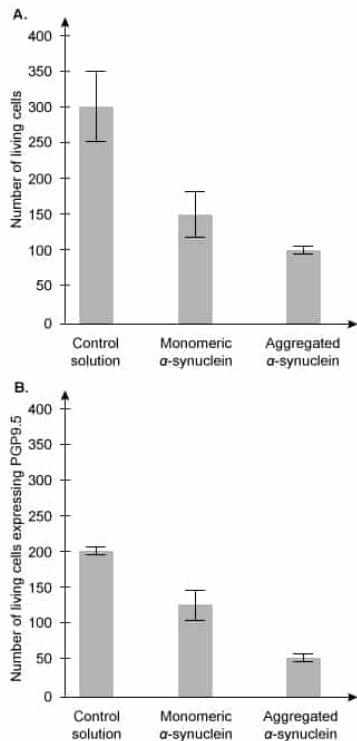


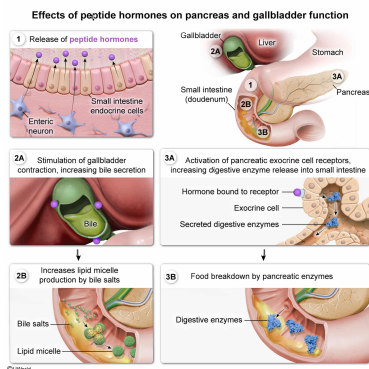
Figure 2 Number of living cells (A) and number of living cells expressing PGP9.5 (B) 5 days after exposure to either a control solution lacking α -synuclein, 0.5 μ M monomeric (nonaggregated) mutant α -synuclein proteins, or 0.5 μ M aggregated mutant α -synuclein proteins

A peptide hormone secreted by small intestine endocrine cells and enteric neurons in WT mice was found to increase the release of digestive enzymes and stored bile into the small intestine lumen but did not increase the secretion of digestive enzymes into the stomach. Given this information, which of the following is NOT likely an effect of this hormone on digestive processes?

- ☐ A. Activation of receptors on exocrine cells of the pancreas [9%]
- ☐ B. Stimulation of gallbladder contraction [5%]
- ☐ C. Increased production of lipid micelles in the small intestine [5%]
- ☒ D. Enhanced release of pepsinogen from chief cells [79%]

Correct
79% answered correctly

Explanation:



The **endocrine system** modulates physiological activities by secreting **hormones** into the bloodstream that **travel** via the circulatory system to target tissues. These **hormones** may bind extracellular or intracellular receptors on target cells (eg, **pancreatic cells**), initiating a cellular response. In addition to major **endocrine glands**, endocrine cells in the **gastrointestinal system** also secrete certain hormones that regulate **digestive processes**. For example, specialized stomach and small intestine cells secrete a hormone called gastrin that signals for increased hydrochloric acid (HCl) secretion from parietal cells in the stomach. In contrast, another hormone released by the small intestine, secretin, inhibits gastric acid secretion.

This question states that a peptide hormone secreted by small intestine endocrine cells and enteric neurons in WT mice was found to **increase** the release of digestive enzymes and stored bile into the **small intestine lumen**.

Therefore, this peptide hormone would increase the flow of digestive enzymes as follows:

- **Activation of receptors** on exocrine cells of the **pancreas**. This causes the pancreas to secrete **proteolytic** digestive enzymes (eg, trypsinogen, chymotrypsinogen) into the duodenum of the small intestine (**Choice A**).

In addition, this peptide hormone would increase the flow of stored bile into the small intestine lumen as follows, and also affect production of lipid micelles:

- **Stimulation of gallbladder contraction**. This causes the gallbladder to release its stores of nonenzymatic bile, which is originally produced by the **liver** (**Choice B**).
- **Increased the production of small lipid micelles** in the small intestine. Bile released by the gallbladder, which contains bile salts, bile pigments (eg, bilirubin), and cholesterol, participates in the digestion of fats in the small intestine via **emulsification**. Specifically, bile salts have both a hydrophilic region that can associate with water and a hydrophobic region that can associate with the surface of lipids. This allows bile salts to act as detergents and break down large lipid globules into **smaller droplets** called **micelles** (**Choice C**).

However, this hormone did *not* increase the secretion of digestive enzymes into the stomach.

Chief cells release the digestive enzyme pepsinogen into the stomach, where its active form (ie, pepsin) aids in protein digestion. However, the question states that the peptide hormone did *not* increase (ie, enhance) the secretion of digestive enzymes into the stomach. Therefore, enhanced release of pepsinogen from chief cells is *not* a likely effect of this hormone on digestive processes.

Educational objective:

Certain hormones secreted by endocrine cells in the gastrointestinal system regulate digestive processes, including the secretion of digestive enzymes and release of bile.

Time spent: 0 sec
Subject:
Biology

QID: 403226
System:
Digestion and Excretion

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Experiment 1

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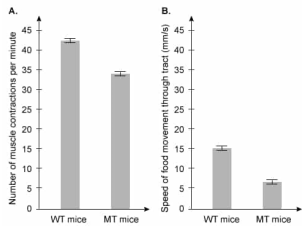


Figure 1 Number of muscle contractions (A) and speed of material movement through the tract (B) in small intestines of WT and MT mice

Experiment 2

Equal numbers of cells (approximately 400) from the myenteric nerve plexus were isolated from WT mice and exposed to either a control solution lacking α -synuclein, 0.5 μ M monomeric (nonaggregated) mutant α -synuclein proteins, or 0.5 μ M aggregated mutant α -synuclein proteins. After 5 days, the total numbers of living cells were determined along with the numbers of these living cells expressing a protein called PGP9.5 (Figure 2), which is known to be expressed in healthy neurons.

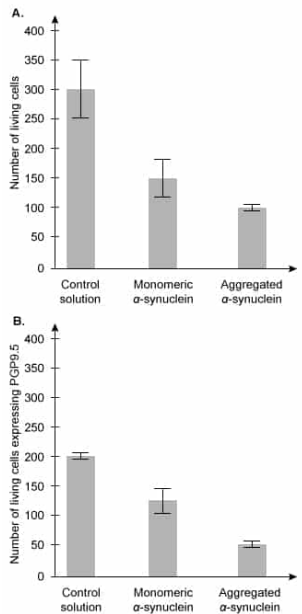
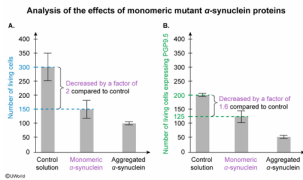


Figure 2 Number of living cells (A) and number of living cells expressing PGP9.5 (B) 5 days after exposure to either a control solution lacking α -synuclein, 0.5 μ M monomeric (nonaggregated) mutant α -synuclein proteins, or 0.5 μ M aggregated mutant α -synuclein proteins

Compared to the control solution, monomeric mutant α -synuclein proteins decreased the total number of living cells and the number of living cells expressing PGP9.5 (Figure 2) by factors of approximately:

- ☐ A. 1.6 and 4, respectively. [5%]
- ☒ B. 2 and 1.6, respectively. [87%]
- ☐ C. 2 and 3, respectively. [5%]
- ☐ D. 3 and 4, respectively. [1%]

Explanation:



The **enteric nervous system** (ENS), consisting of both the submucosal nerve plexus and the myenteric nerve plexus, innervates the gastrointestinal tract and regulates digestive function. Although the ENS does not require input from the **central nervous system** (CNS), the **autonomic nervous system** can modulate ENS function.

In Experiment 2 described in the passage, equal numbers of cells (approximately 400) from the myenteric nerve plexus were isolated from WT mice and exposed to either a control solution (ie, no α -synuclein), 0.5 μ M monomeric (nonaggregated) mutant α -synuclein proteins, or 0.5 μ M aggregated mutant α -synuclein proteins. Figure 2 shows the number of living cells and the number of these living cells expressing a protein called PGP9.5 five days after each treatment. PGP9.5 is known to be expressed in healthy **neurons**.

This question asks for a comparison of the effects of the control solution and **monomeric mutant α -synuclein proteins** on these myenteric nerve plexus cells. Figure 2A shows that there were approximately **300** living cells present after treatment with the control solution and approximately **150** living cells present after treatment with monomeric mutant α -synuclein proteins. Therefore, the factor by which monomeric mutant α -synuclein proteins **decreased** the number of living cells compared to the control solution can be calculated as follows:

$$\begin{aligned} &\text{Factor by which monomeric mutant } \alpha\text{-synuclein proteins decreased number of living cells} \\ &= \frac{\text{Number of living cells after control solution treatment}}{\text{Number of living cells after monomeric mutant } \alpha\text{-synuclein treatment}} = \frac{300}{150} = 2 \end{aligned}$$

Figure 2B shows that approximately **200** living cells expressed PGP9.5 after treatment with the control solution and that approximately **125** living cells expressed PGP9.5 after treatment with monomeric mutant α -synuclein proteins. Therefore, the factor by which monomeric mutant α -synuclein proteins **decreased** the number of living cells expressing PGP9.5 compared to the control solution can be calculated as follows:

$$\begin{aligned} &\text{Factor by which monomeric mutant } \alpha\text{-synuclein proteins decreased number of living cells expressing PGP9.5} \\ &= \frac{\text{Number of living cells expressing PGP9.5 after control solution treatment}}{\text{Number of living cells expressing PGP9.5 after monomeric mutant } \alpha\text{-synuclein treatment}} = \frac{200}{125} = 1.6 \end{aligned}$$

Only **Choice B** shows this result.

(Choices A, C, and D) These values result from *incorrect* analysis of the data.

Educational objective:

A specialized division of the nervous system, called the enteric nervous system (ENS), innervates and regulates the function of the gastrointestinal tract. In addition, ENS functions may be modulated by the autonomic nervous system.

Time spent: 0 sec
Subject:
Biology

QID: 403227
System:
Digestion and Excretion

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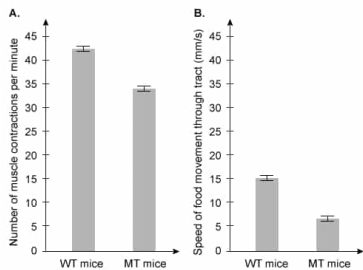


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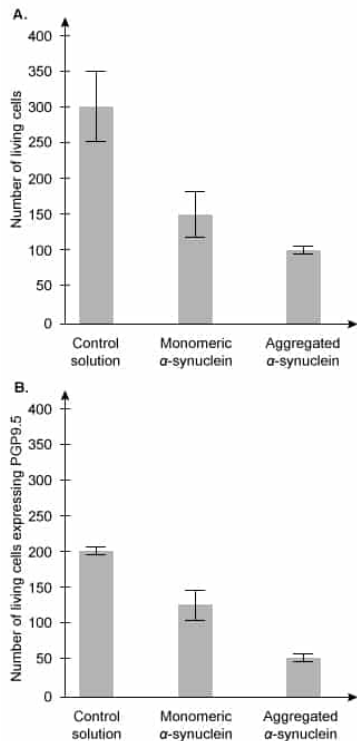


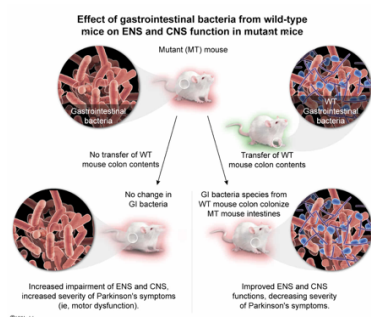
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In a different experiment, the scientists found that transferring contents from the colons of WT mice into the colons of MT mice decreased the severity of PD symptoms that developed in MT mice over time. The best explanation for this observation is that:

- ☐ A. healthy bacterial species from WT mice transferred to the colons of MT mice could express large amounts of wild-type α -synuclein protein and restore ENS and CNS function. [48%]
- ☐ B. many antimicrobial compounds in the GI tracts of WT mice significantly decreased GI bacteria numbers in MT mice, enhancing neuron function. [4%]
- ☒ C. WT mice contained specific amounts and types of GI bacterial species that could colonize the intestines of MT mice and improve ENS and CNS function. [43%]
- ☐ D. the GI tracts of WT mice contained mostly bacterial species with genetic similarities, increasing bacterial genetic diversity and neuron function in MT mice. [3%]

Incorrect. Correct answer is C
43% answered correctly

Explanation:



The **large intestine** contains a diverse array of **bacterial** species (ie, gut flora) that aid in digestive processes. For example, some of these bacteria can process food items that would otherwise be indigestible, such as certain carbohydrates. Gut bacteria metabolize these undigested carbohydrates into short-chain fatty acids that can then be absorbed and used by the body for energy. In addition, some bacterial species in the large intestine synthesize certain vitamins.

In this question, a different experiment is described in which contents from the **colons** (ie, a large **intestine segment**) of wild-type (WT) mice were transferred into the colons of mutant (MT) mice (ie, mice with the α -synuclein mutation). The scientists found that this procedure **decreased** the severity of Parkinson's symptoms that developed in the MT mice over time.

Because the colon contents of WT mice would contain a portion of their gastrointestinal (GI) bacterial species, this procedure most likely supplied MT mice with gut flora from healthy WT mice. Therefore, the best explanation for decreased severity of Parkinson's symptoms in MT mice is that **WT mice** contained **specific amounts and types of GI bacteria** that could **colonize** the intestines of MT mice and **improve ENS and CNS** function.

(Choice A) According to the passage, α -synuclein proteins are **expressed** by mice, *not* by GI bacteria. Therefore, healthy *bacterial* species would *not* likely have expressed large amounts of wild-type α -synuclein.

(Choice B) Because the large intestine contains many types of GI bacterial species, it is *unlikely* that the GI tracts of WT mice contained many antimicrobial compounds (ie, compounds that *kill* bacteria). In addition, because GI bacteria are key to digestive health, a significant *decrease* in bacterial numbers in the intestines of MT mice would have been *unlikely* to *enhance* the function of neurons involved in digestion.

(Choice D) GI bacterial species are generally very *diverse*, allowing them to carry out various functions in the large intestine. Therefore, GI bacteria from WT mice likely had many genetic differences (*not* similarities) that served to *increase* bacterial genetic diversity in MT mice.

Educational objective:

Diverse bacterial species in the large intestines of many species (eg, humans, mice) perform functions important to digestive health, including the digestion of certain materials that would otherwise be indigestible as well as vitamin synthesis.

Time spent: 0 sec
Subject:
Biology

QID: 403228
System:
Digestion and Excretion

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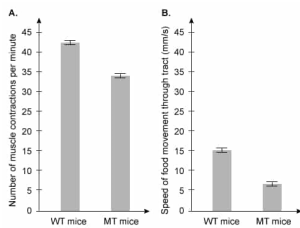


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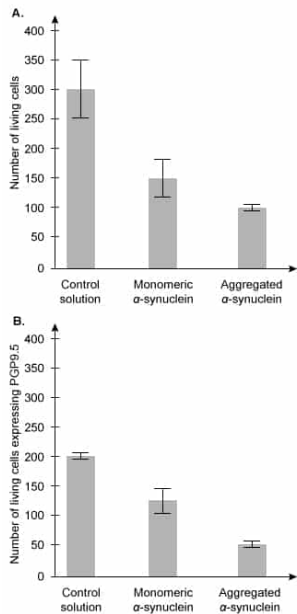


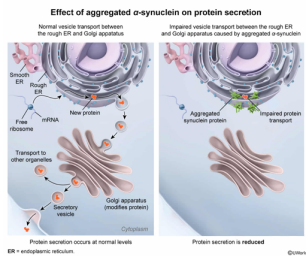
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Which statement most likely explains the reduction in protein secretion observed in the motor neurons of individuals with PD?

- ☐ A. Aggregated α -synuclein proteins promote higher expression of cytoskeletal proteins in the axons of motor neurons. [2%]
- ☐ B. Aggregated α -synuclein proteins increase the production of molecules that negatively regulate smooth endoplasmic reticulum function. [6%]
- ☐ C. Aggregated α -synuclein proteins reduce expression of plasma membrane proteins required for the facilitated diffusion of molecules from motor neurons. [31%]
- ☒ D. Aggregated α -synuclein proteins impair intracellular vesicle transport between the rough endoplasmic reticulum and the Golgi apparatus in motor neurons. [59%]

Correct
58% answered correctly

Explanation:



The **endoplasmic reticulum** is a cytosolic **organelle** composed of a network of membranous sacs and tubules. It is divided into the **rough endoplasmic reticulum** (RER) and the smooth endoplasmic reticulum (SER).

The RER has ribosomes located along its surface that **translate** proteins destined for the **secretory pathway**. Translated proteins may undergo certain **modifications** (eg, glycosylation) in the RER. Next, proteins in the RER are transported to the **Golgi apparatus**, where additional modifications may occur followed by the packaging of the proteins into vesicles. Proteins transported via vesicle from the Golgi apparatus may be secreted from the cell, added to the cell membrane, or used by certain organelles (eg, **lysosomes**).

As the passage states, individuals with Parkinson's disease (PD) exhibit the accumulation of aggregated α-synuclein proteins (ie, Lewy bodies) in the **central nervous system** (CNS). Lewy bodies may **impair** motor neuron function by decreasing protein secretion from these neurons.

Because proteins secreted from cells are synthesized in the RER and transported to the Golgi apparatus for packaging, the most likely explanation for this **reduced** protein secretion is that aggregated α-synuclein proteins **impair intracellular vesicle transport** between the **RER** and **Golgi apparatus** in motor neurons.

(Choice A) Microtubules, components of the **eukaryotic cytoskeleton**, have several functions, including serving as tracks for the **intracellular transport** of vesicles by motor proteins (ie, kinesin, dynein). Accordingly, **higher** expression of microtubules in the **axons** of motor neurons would likely facilitate (*not* decrease) transport and secretion of protein-containing vesicles.

(Choice B) The lumen of the SER contains enzymes that synthesize **lipids**, including cholesterol and cholesterol-derived molecules (eg, steroid hormones). However, the SER contains no ribosomes and therefore does *not* participate in protein synthesis. Therefore, negative regulation of SER function would *not* explain reduced protein secretion.

(Choice C) **Facilitated diffusion**, a form of passive transport (ie, does *not* require ATP), allows **small** molecules (eg, polar solutes) to cross the plasma membrane via transport proteins. However, much **larger** molecules (eg, proteins) are generally transported across the membrane via **exocytosis**, a form of **membrane transport** that *does* require energy.

Educational objective:

Proteins synthesized by ribosomes on the rough endoplasmic reticulum are transported to the Golgi apparatus. In the Golgi apparatus, proteins are modified and packaged into vesicles to be transported to the cell membrane, used in organelles such as lysosomes, or secreted out of the cell.

Time spent:0 sec	QID:403229
Subject:	System:
Biology	Digestion and Excretion